

DANIEL SZUREK

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Tuscaloosa, AL, USA

EDUCATION

The University of Alabama

Master of Science in Computer Science
Accelerated Master's Program (AMP)

May 2025 - December 2026

GPA: 3.67

The University of Alabama

Bachelor of Science in Computer Science
Minor in Mathematics, Honors College, Upsilon Pi Epsilon

August 2022 - May 2026

GPA: 3.87

PROFESSIONAL EXPERIENCE

General Motors

ADAS Software Engineer Intern

Milford, MI, USA

May 2026 - August 2026

- Engineered an agentic LLM-driven automation pipeline in Python using the GitHub API to auto-generate signal recording definition functions, reducing manual authoring time by 80%
- Developed data recording infrastructure supporting over 5 ADAS feature teams, ensuring reliable event-data capture for safety-critical validation across thousands of test miles
- Supported on-vehicle data collection at the Milford Proving Grounds, debugging recording configurations against live test runs to match feature-team signal requirements
- Collaborated with ADAS feature teams to translate recording requirements into automated, scalable software solutions

The University of Alabama EcoCAR

Connected and Automated Vehicle (CAV) Lead

Tuscaloosa, AL, USA

September 2025 - Present

- Architected the full autonomous navigation stack on a pre-production Cadillac Lyriq, integrating Lane Centering Control, Cooperative Adaptive Cruise Control, and V2X connectivity with stock onboard systems
- Developed and tuned ADAS control algorithms in MATLAB/Simulink including an MPC/PID longitudinal controller, Stanley lateral controller, and GNN sensor fusion pipeline, cutting lateral tracking error to under 30 cm in HIL testing
- Validated safety-critical performance through Hardware-in-the-Loop (HIL) and Vehicle-in-the-Loop (VIL) testing, building automated test cases that caught controller faults before on-vehicle deployment
- Led a 15-person team in GM's EcoCAR EV Challenge, earning 1st in Vehicle Technical Specifications, 2nd in MathWorks Model-Based Design, and 2nd in CAV Presentations

SSAB

Data Science Co-Op

Mobile, AL, USA

May 2025 - August 2025

- Engineered automated ETL pipelines in Azure Data Factory and SQL to process large-scale manufacturing datasets, streamlining data flow across production systems
- Designed Power BI dashboards automating quarterly financial reporting and production capacity modeling, enabling leadership to set targets from live sensor data
- Built and deployed cost comparison tools in Tableau and SQL, simplifying workflows for high-volume manufacturing datasets
- Presented data-driven insights to senior leadership and facilitated international cross-functional meetings with colleagues in Europe

PROJECTS

Driveability Accessible Driving Resource

[Link to Project](#)

- Built a social-media web platform with resources and reviews on assisted-driving tools for disabled users, completed as my degree capstone project

Gaussian Process and MPC Controller for Automatic Intersection Navigation via V2X

[Link to Project](#)

- Developed a custom controller for Automatic Intersection Navigation that accounts for SPaT V2X data loss from radio noise via a trained Gaussian Process

Super-Resolution Facial Recognition Pipeline

[Link to Project](#)

- Developed a custom Convolutional Neural Network (CNN) pipeline to perform facial recognition on low-resolution inputs
- Implemented a Deep Super-Resolution (DSR) model to upscale images before feeding them into a quantized EdgeFace model, improving accuracy on degraded inputs
- Applied model quantization techniques to reduce model size for efficient deployment without significant accuracy loss

TECHNICAL SKILLS

ADAS & Autonomy: ROS 2, Simulink, RTMaps, Model Predictive Control (MPC), Stanley Controller, Sensor Fusion (GNN), Hardware-in-the-Loop (HIL), Vehicle-in-the-Loop (VIL), V2X / DSRC, dSPACE AUTERA

AI & ML: PyTorch, Computer Vision (CNNs), Gaussian Processes, Reinforcement Learning, LLM Integration (OpenAI API), LangChain

Cloud & DevOps: Azure (Data Factory, Databricks), Docker, Git, Linux (Ubuntu)

Languages: Python, C/C++, MATLAB, SQL, Bash, Flutter, JavaScript, Swift, TypeScript, HTML